

FPGA-Based Multi-Joint Robotic Arm Control Using Verilog HDL and PWM Technique

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Abstract

In this research, the design, development and experimental testing of an FPGA based multi-joint robot arm controlled by Verilog HDL have been done. A PWM generator has been designed and simulated on a Nexys-4 FPGA board to generate variable duty cycles. In order to test the performance of the PWM controller, it was initially used to control the speed and torque of both a DC motor and a DC geared motor using an L293D H-Bridge driver at different duty cycles. Using the experimental observations from the tests, the PWM signal has been designed for SG5010 servo motors for accurate positioning. A three joint robot arm including shoulder, elbow and wrist has been designed using finite state machine (FSM). **Keywords:** FPGA, Verilog HDL, Robotic Arm, Pulse Width Modulation (PWM)

Literature review and Theoretical Background

FPGA-based control has become increasingly popular in robotics due to its deterministic operation, hardware parallelism, and ability to control multiple actuators simultaneously. Mansoor, Khalil, and Jasim [1] demonstrated FPGA-based position control of DC servo motors using a soft-core processor for robotic arm motion. Kung et al. [2] proposed an FPGA-based servo control architecture for a six-axis articulated manipulator, highlighting the benefits of parallel hardware implementation for coordinated multi-axis control. An FPGA-based 5 degrees of freedom mobile robotic arm integrating control of both DC and servo motors was designed by Karci and Tangel [3]. Wan et al. [4] provided a review on robotic computing using FPGA, highlighting low latency, deterministic processing, and power efficiency. In view of these previous works, the current study uses a step-by-step approach in which DC motors, geared DC motors, and servos are first considered before designing a 3 joint robotic arm. This design employs Pulse Width Modulation (PWM), where duty cycle is expressed as

$$D = \frac{T_{ON}}{T} \times 100\% \quad (1)$$

Average voltage to the motor is $V_{avg} = D \times V_s$, meaning higher duty cycle translates to higher average voltage and hence increased motor speed when loads are equal. PWM is used to control the DC and geared DC motors, whereas calibrated pulse width is used to control the positions of SG5010 servos.

Results and Discussion

PWM Verification Using LEDs

Prior to connecting the motors to the FPGA, the PWM signal generator was tested using the LEDs on the Nexys-4 FPGA board. The PWM output signal was then connected to one of the LEDs, and the duty cycle was modified with different inputs. As the duty cycle increased, the LED appeared brighter due to the increased-ON time within each PWM period, while lower duty cycles resulted in reduced brightness. This simple hardware verification confirmed the correct generation of variable-duty-cycle PWM signals and validated the functionality of the Verilog HDL design before applying the PWM output to the L293D motor driver for DC and geared DC motor speed control.



Fig 1: LED Brightness for 20% duty cycle at 00



Fig 2: LED Brightness for 50% duty cycle at 01



Fig 3: LED Brightness for 80% duty cycle at 10



Fig 4: LED Brightness for 100% duty cycle at 11

DC Motor Speed Analysis

Table 1: DC Motor Observation

Duty Cycle (%)	Motor Response
20	Slow (Fig 7.5) – 640-750 RPM
50	Moderate (Fig 7.6) – 900-990 RPM
80	Fast (Fig 7.7) – 1000-1200 RPM
100	Maximum Speed (Fig 7.8) – 1250-1300 RPM

The DC motor exhibited a nearly proportional increase in speed with increasing PWM duty cycle. Since the average voltage supplied to the motor increases with duty cycle, the motor rotates faster at higher duty cycles. The FPGA consistently produced stable PWM signals throughout the experiment, demonstrating that PWM is an effective technique for controlling the speed of DC motors..



Fig 5: Speed of DC motor in RPM for 20% duty cycle



Fig 6: Speed of DC motor in RPM for 50% duty cycle



Fig 7: Speed of DC motor in RPM for 80% duty cycle



Fig 8: Speed of DC motor in RPM for 100% duty cycle

DC Geared Motor Speed Analysis

Table 2: DC Geared Motor Observation

Duty Cycle (%)	Motor Response
10	No Rotation
20	No Rotation
31.5	Slow Rotation (Fig 7.9) – 15-25 RPM
50	Stable Rotation (Fig 7.10) – 225-235 RPM
80	High Torque (Fig 7.11) – 250-255 RPM
100	Maximum Speed (Fig 7.12) – 265-270 RPM

The geared DC motor behaved differently from a conventional DC motor because it required a higher PWM duty cycle before it could begin rotating. This is due to the gearbox, which increases the torque needed during startup while reducing the motor's output speed. Although the geared motor operated at a slower speed, it produced much greater torque, making it more suitable for robotic applications that require lifting or moving heavier loads.



Fig 9: Speed of DC Geared motor in RPM for 31.5% duty cycle



Fig 10: Speed of DC Geared motor in RPM for 50% duty cycle



Fig 11: Speed of DC Geared motor in RPM for 80% duty cycle



Fig 12: Speed of DC Geared motor in RPM for 100% duty cycle

Three-Joint Robotic Arm Movement

Table 3: Home Position of all the servo motors

Joint	Initial Angle
Shoulder	0°
Elbow	30°
Wrist	15°

The robotic arm used MG995 and SG5010 servos to perform the operations of shoulder, elbow, and wrist joints respectively. All the three servos were supplied with PWM pulses through the FPGA which allowed for precise positioning of angles with respect to predefined combinations of switches as shown in Table I. All the recorded joint movements were accurate and there was no instability in the process of positioning the joints as expected. Different inputs from switches caused different configurations of the robot arm indicating proper implementation of pulse widths and coordination of all joints. The finite state machine allowed performing the required movements in a sequential way. **Table 4:** Predefined robotic arm poses and corresponding joint angles for different FPGA switch combinations

Switch (SW[2:0])	Pose	Shoulder (°)	Elbow (°)	Wrist (°)	Description
000	Home Position	0°	30°	15°	Default position after reset and after every motion cycle
001	Pick Position	45°	60°	30°	Moves arm toward the object for picking
010	Lift Position	90°	90°	45°	Lifts the object vertically
011	Place Position	135°	60°	90°	Places the object at the target location
100	Extended Reach	180°	90°	45°	Maximum forward extension of the arm
101	Inspection Position	90°	45°	135°	Demonstrates an alternate inspection/observation pose
110	Rest Position	45°	30°	15°	Intermediate standby position
111	Reset/Home	0°	30°	15°	Returns the robotic arm to the home position

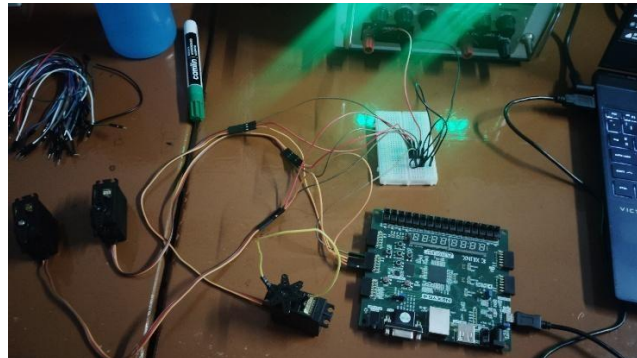


Fig 13: Experimental set up of all the servo motors used for 3 joints

Conclusion

This internship concentrated on the design, development, and experimental testing of an FPGA-based robotic arm controller utilizing the Nexys-4 FPGA platform and Verilog HDL programming language. A PWM generator was designed and tested using the on-board LEDs prior to being used to control a DC motor as well as a DC geared motor by means of an L293D motor controller in order to investigate the impact of duty cycle on the speed of the motor. The project was further applied to SG5010 servo motors, where calibrated PWM pulse lengths were used to position the joints accurately.

References

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